Coin Change

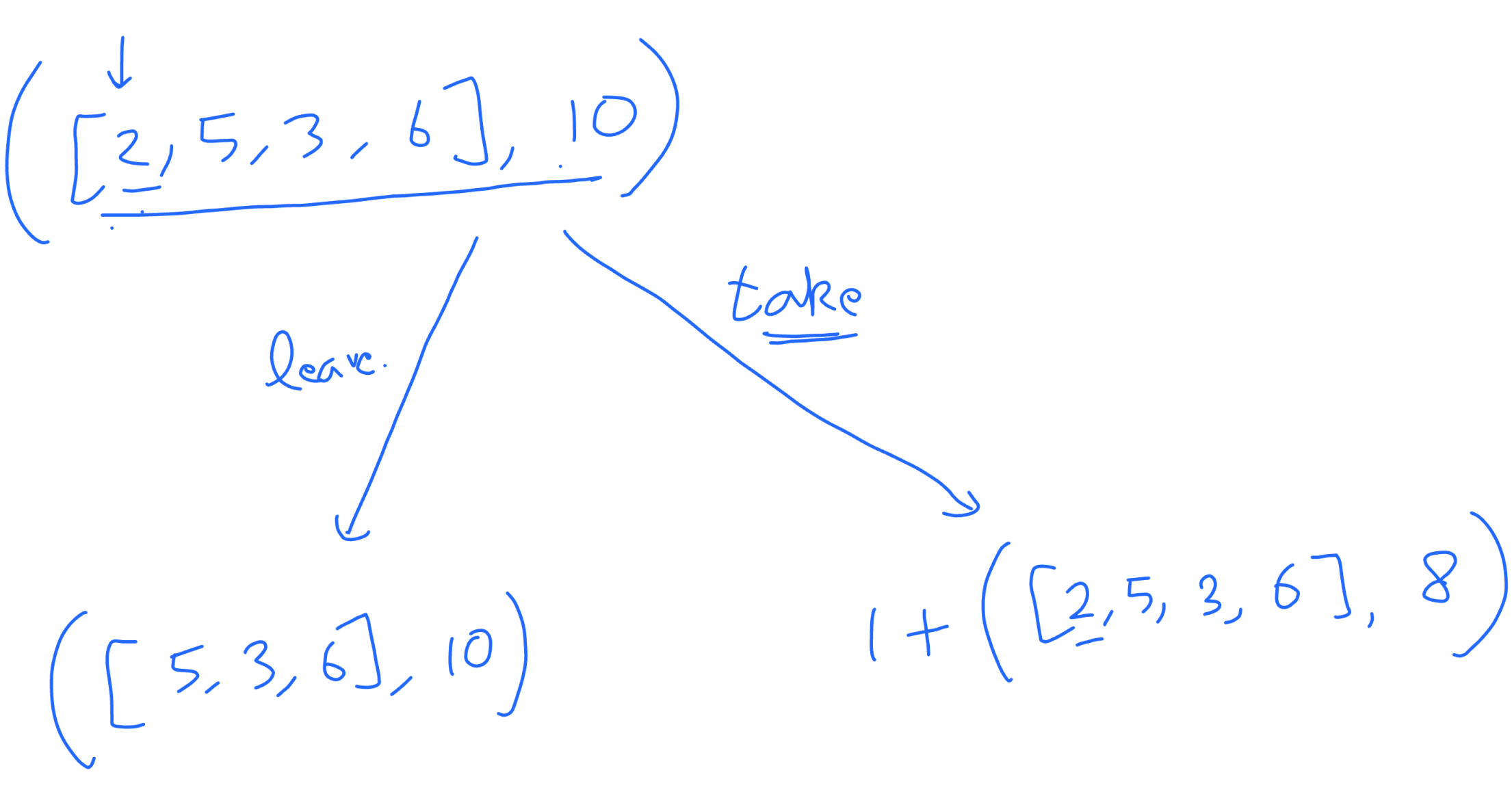
value = 5
coins = [3, 6, 3].
o/p = -1

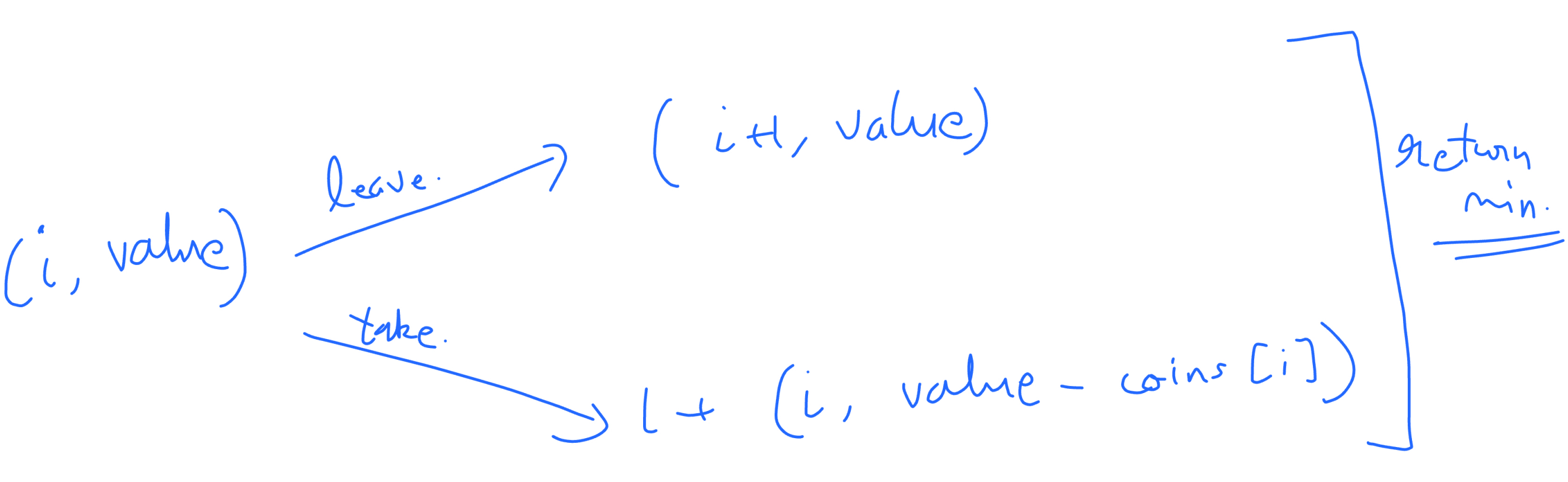
value = 10
coins = [2, 5, 3, 6].
o/p = 2   (5 + 5 = 10)

[1, 2, 3, 5, 6].

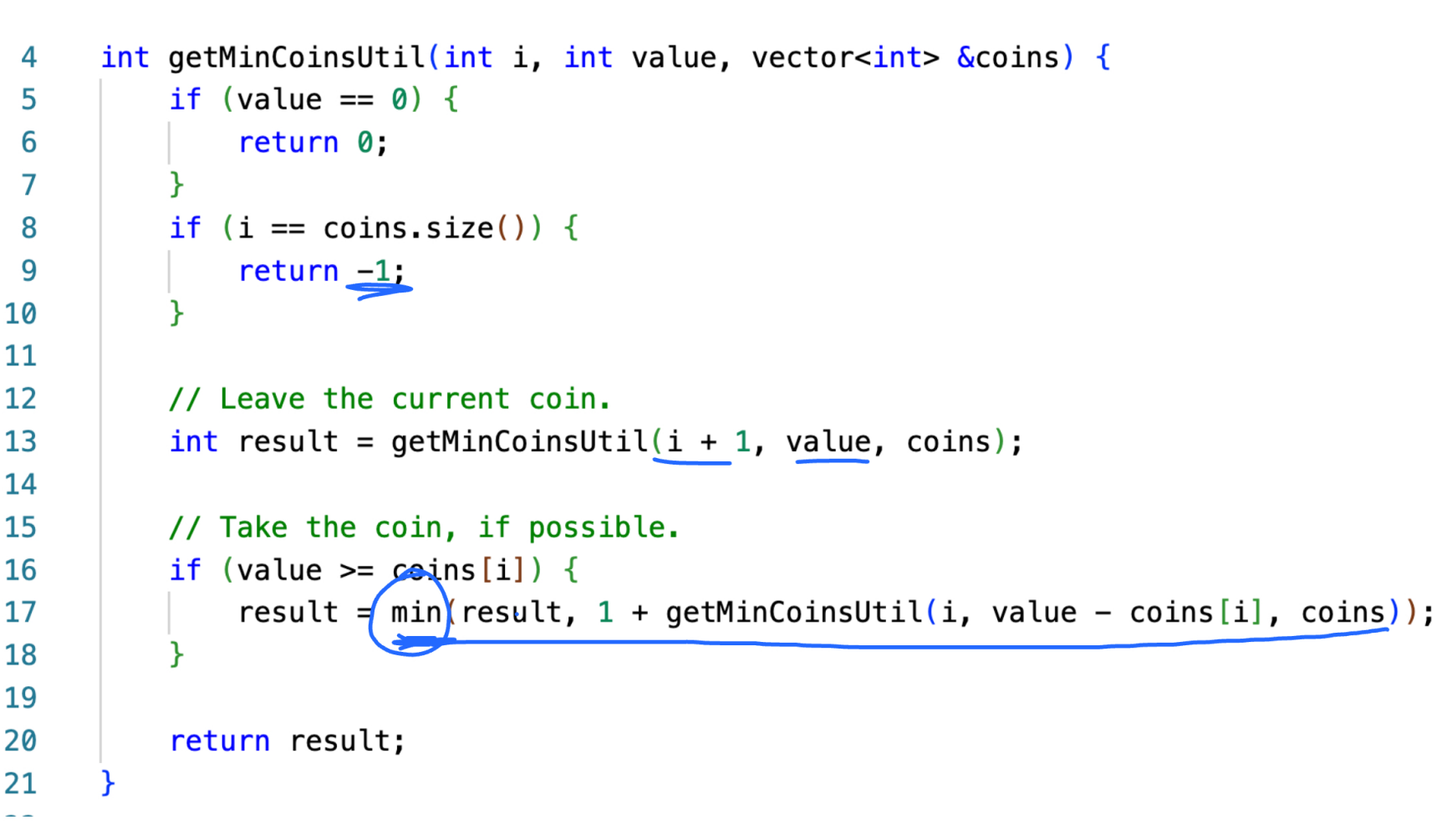
value = ~~10~~ ~~4~~ ~~1~~ 0

res = ~~0~~ ~~1~~ ~~2~~ 3
→ (6 + 3 + 1) X
(5 + 5) ✓


([2, 5, 3, 6], 10)
leave
take
([5, 3, 6], 10)
1 + ([2, 5, 3, 6], 8)

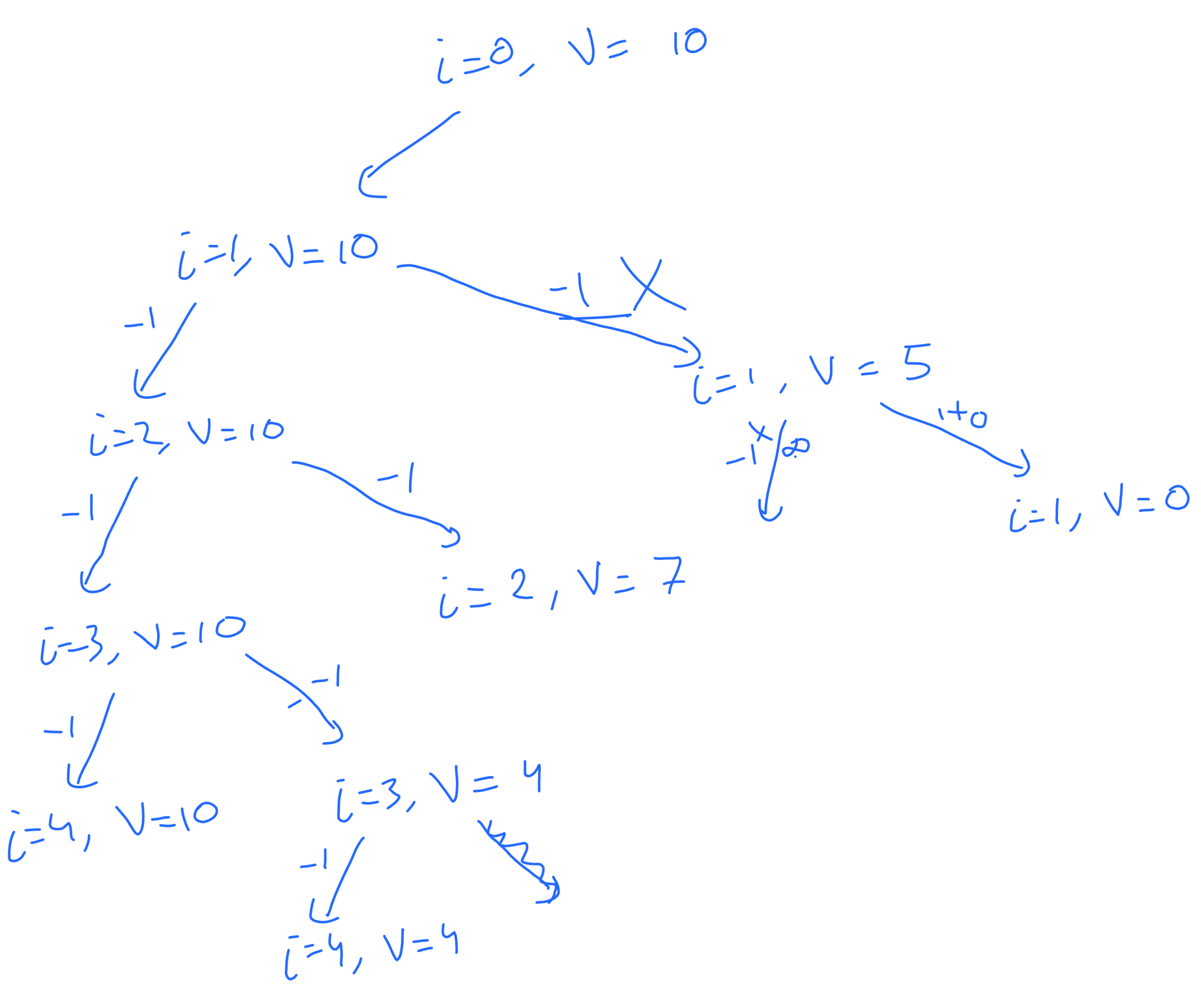

(i, value)
leave
(i+1, value)
take
1 + (i, value - coins[i])
return min.

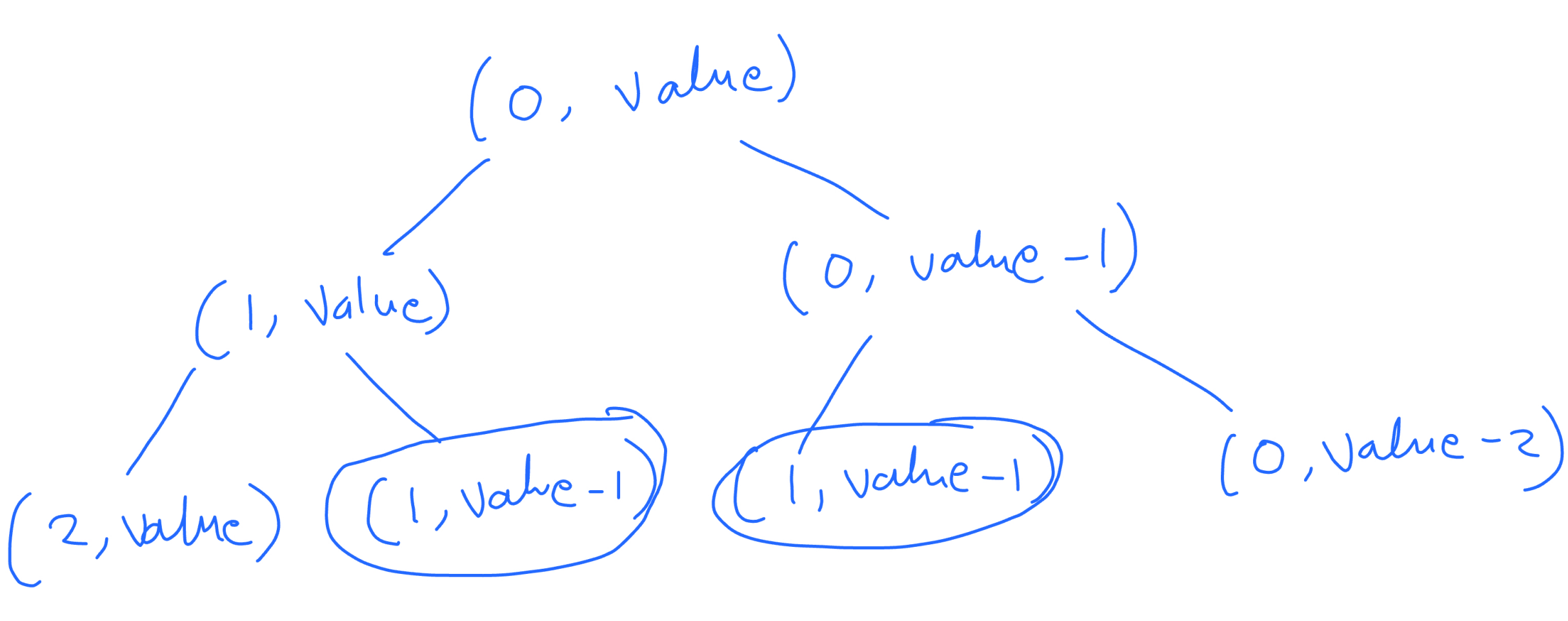
50   10



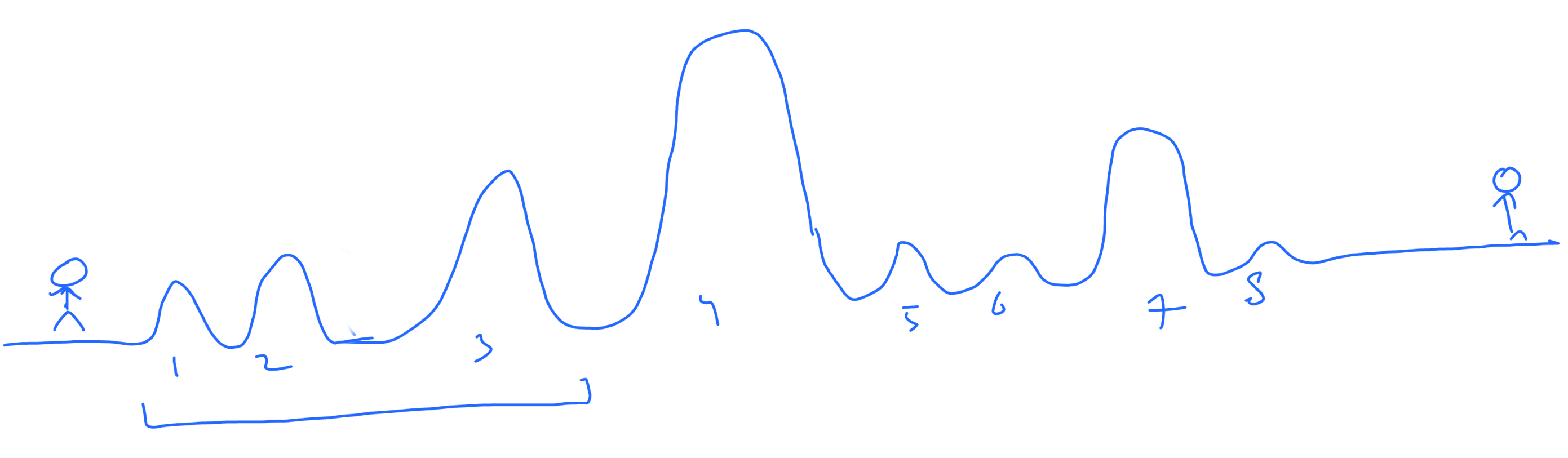
```
int getMinCoinsUtil(int i, int value, vector<int> &coins) {
    if (value == 0) {
        return 0;
    }
    if (i == coins.size()) {
        return -1;
    }

    // Leave the current coin.
    int result = getMinCoinsUtil(i + 1, value, coins);

    // Take the coin, if possible.
    if (value >= coins[i]) {
        result = min(result, 1 + getMinCoinsUtil(i, value - coins[i], coins));
    }

    return result;
}
```

0 1 2 3.
[2, 5, 3, 6].


i=0, V= 10
i=1, V=10
-1
i=2, V=10
-1
i=3, V=10
-1
i=4, V=10
-1
i=2, V=7
-1
i=3, V=4
-1
i=4, V=4
-1 X
i=1, V=5
-1/∞
1+0
i=1, V=0


(0, value)
(1, value)
(0, value -1)
(2, value)
(1, value-1)
(1, value-1)
(0, value -2)

Greedy.



[4, 2, 3, 6, 8]   T = 10

o/p : 3   (2+3+4)

[2, 3, 4, 6, 8],   T = ~~10~~ ~~8~~ ~~5~~ 1

+&3

[Fractional Knapsack]

vals = [60 100 120].
wts = [10 20 30].
W = ~~50~~ ~~20~~ 0

value = 120 + 100 = 120.

Greedily picking the items with the greatest value first.
X

vals = [60 100 120]
wts = [10 20 30]
W = ~~50~~ ~~40~~ ~~20~~ 0

value = 60 + 100 + $\frac{2}{3}$(120)
= 60 + 100 + 80
= 240.

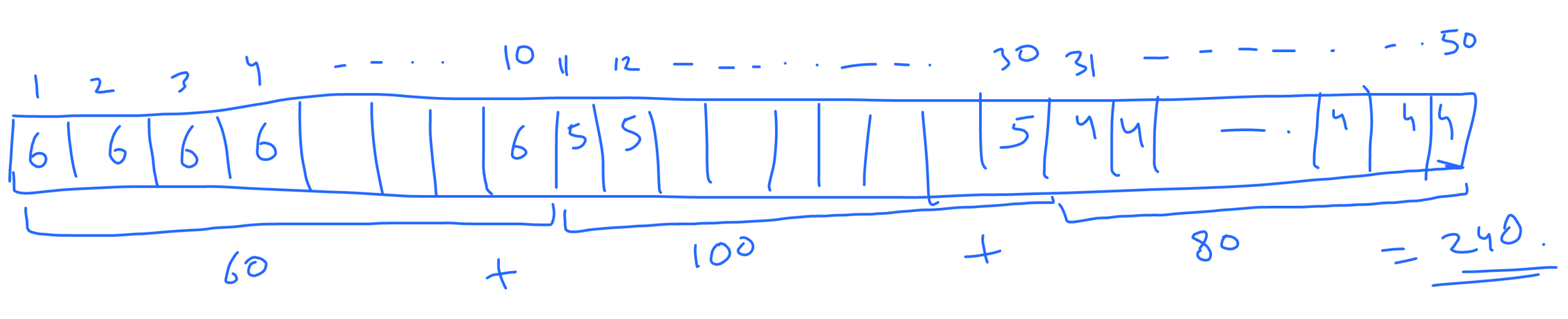
X Greedily picking the items with smallest wt. first.

vals = [6 10 120]
wts = [10 20 30]
W = ~~50~~ ~~40~~ ~~20~~ 0

value = 6 + 10 + 80
= 96.

vals = [60 100 120]
wts = [10 20 30]
W = 50

6 5 4


1 2 3 4 - - - . . 10 11 12 - - - . . - - . 30 31 - - - - . . . 50
6 6 6 6 6 5 5 5 4 4 4 4 4
60
+
100
+
80
= 240.

Greedily consider the items with the largest value per unit wt. ✓

val, wt, $\frac{val}{wt}$

{val, wt}.   {$\frac{val}{wt}$, wt}.

vals = [60 100 120]
wts = [10 20 30]
W = 50
{(6, 10) (5, 20) (4, 30)}.

W = ~~50~~ ~~40~~ ~~20~~ 0

res = ~~0~~ ~~60~~ ~~160~~ 240

6.66 5.26 4.13
vals = [60 100 120]
wts = [9 19 29]
W = ~~50~~ ~~41~~ ~~22~~ 0

res = ~~60~~ 160 + $\frac{22}{29}$(120)
160 + 91.03
= 251.03.

$\frac{int}{int}$ → int | $\frac{int}{double}$ OR $\frac{double}{int}$ → double

---

Job Sequencing

id deadline profits.
→ (1, 4, 20)
(2, 1, 10)
→ (3, 1, 40)
→ (4, 1, 30)

0-1   1-2   2-3   3-4   4-5 . . . .
3   1
(40)   (20)
= 60.

→ (1 2 100)
(2 1 19)
(3 2 27)
(4 1 25)
(5 1 15)

0-1 1-2
4 1 → 125 X
0-1 1-2
1 3 → 127 ✓

→ (1 2 100)
(2 1 19)
→ (3 1 27)
(4 1 25)
(5 1 15)

0-1 1-2
(3) (1)
27 100 → 127.